

# Experiment 1.2: Design and Performance Evaluation of a Butterworth IIR High-Pass Filter

## MATLAB Code:

```
%% Experiment 1.2: Butterworth IIR High-Pass Filter
% Design and Performance Evaluation

clc;
clear;
close all;

%% PARAMETERS
Fs = 8000;
Fp = 3000;
Fst = 1000;
Rp = 1;
Rs = 20;

%% NORMALIZATION
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

%% ORDER CALCULATION
[N, Wn] = buttord(Wp, Ws, Rp, Rs);

%% FILTER DESIGN
[b, a] = butter(N, Wn, 'high');

fprintf('Filter Order = %d\n', N);
fprintf('Cutoff Frequency = %.4f\n', Wn);

%% CREATE SINGLE FIGURE
figure;

%% 1. Magnitude Response
[H, w] = freqz(b, a, 512);
subplot(3,2,1);
plot(w/pi, abs(H));
title('Magnitude Response');
xlabel('Normalized Frequency');
ylabel('|H(w)|');

%% 2. Phase Response
subplot(3,2,2);
plot(w/pi, angle(H));
title('Phase Response');
xlabel('Normalized Frequency');
ylabel('Phase (radians)');

%% 3. Impulse Response
subplot(3,2,3);
impz(b, a);
title('Impulse Response');

%% 4. Step Response
subplot(3,2,4);
stepz(b, a);
title('Step Response');

%% 5. Group Delay
subplot(3,2,5);
grpdelay(b, a);
title('Group Delay');

%% 6. Pole-Zero Plot
subplot(3,2,6);
zplane(b, a);
title('Pole-Zero Plot');
saveas(gcf, 'Full_Output.png');
```

Output:

